

MA717: Applied Regression and Experimental Data Analysis

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12th Decemeber 2025

```
if(!require(leaps)) install.packages("leaps")
```

```
## Loading required package: leaps
```

```
library(leaps)
```

Task 1: Data reading and simple exploration (25%)

1.1. Read “College.csv” file into R with following command and use dim() and head() to check if you read the data correct. You should report the number of observations and the number of variables. (5 %)

```
#laoding the dataset
mydata<-read.csv("College.csv", header=T, stringsAsFactors=TRUE)

#checking if data is correct
dim(mydata)
```

```
## [1] 775 17
```

```
head(mydata)
```

```
## Private Apps Accept Enroll F.Undergrad P.Undergrad Outstate Room.Board Books
## 1 Yes 1660 1232 721 2885 537 7440 3300 450
## 2 Yes 2186 1924 512 2683 1227 12280 6450 750
## 3 Yes 1428 1097 336 1036 99 11250 3750 400
## 4 Yes 417 349 137 510 63 12960 5450 450
## 5 Yes 193 146 55 249 869 7560 4120 800
## 6 Yes 587 479 158 678 41 13500 3335 500
## Personal PhD Terminal S.F.Ratio perc.alumni Expend Grad.Rate Elite
## 1 2200 70 78 18.1 12 7041 60 No
## 2 1500 29 30 12.2 16 10527 56 No
## 3 1165 53 66 12.9 30 8735 54 No
## 4 875 92 97 7.7 37 19016 59 Yes
## 5 1500 76 72 11.9 2 10922 15 No
## 6 675 67 73 9.4 11 9727 55 No
```

We have 775 observations and 17 variables

1.2. Use your registration number as random seed, generate a random subset of College data with sample size 700, name this new data as mynewdata. Use summary() to output the summarized information about mynewdata. Please report the number of private and public university and the number of Elite university and non-Elite university in this new data. (12 %)

```
#setting up the random seed
set.seed(2501218)
#generating random subset of size 700
index<- sample(775, size = 700)
```

```
mynewdata <- mydata[index,]
#summarised information about mynewdata
summary(mynewdata)
```

```
## Private      Apps      Accept      Enroll
## No :189      Min.      : 81.0      Min.      : 72.0      Min.      : 35.0
## Yes:511      1st Qu.: 773.2      1st Qu.: 595.8      1st Qu.: 239.8
##              Median : 1562.0      Median : 1111.5      Median : 437.0
##              Mean   : 3018.2      Mean   : 2019.5      Mean   : 782.0
##              3rd Qu.: 3709.0      3rd Qu.: 2447.8      3rd Qu.: 905.5
##              Max.   : 48094.0      Max.   : 26330.0      Max.   : 6392.0
## F.Undergrad  P.Undergrad  Outstate  Room.Board
## Min.      : 139.0      Min.      : 1.0      Min.      : 2340      Min.      :1780
## 1st Qu.: 986.8      1st Qu.: 95.0      1st Qu.: 7383      1st Qu.:3600
## Median : 1711.5      Median : 359.0      Median : 9998      Median :4200
## Mean   : 3701.4      Mean   : 852.6      Mean   :10479      Mean   :4374
## 3rd Qu.: 4139.8      3rd Qu.: 973.0      3rd Qu.:12952      3rd Qu.:5091
## Max.   :31643.0      Max.   :21836.0      Max.   :21700      Max.   :8124
## Books      Personal      PhD      Terminal
## Min.      : 96.0      Min.      : 250      Min.      : 8.00      Min.      : 24.00
## 1st Qu.: 475.8      1st Qu.: 865      1st Qu.: 62.00      1st Qu.: 71.00
## Median : 502.0      Median :1200      Median : 75.00      Median : 82.00
## Mean   : 550.0      Mean   :1342      Mean   : 72.61      Mean   : 79.58
## 3rd Qu.: 600.0      3rd Qu.:1700      3rd Qu.: 85.00      3rd Qu.: 92.00
## Max.   :2340.0      Max.   :6800      Max.   :100.00      Max.   :100.00
## S.F.Ratio  perc.alumni  Expend      Grad.Rate  Elite
## Min.      : 2.50      Min.      : 0.00      Min.      : 3186      Min.      : 10.0      No :631
## 1st Qu.:11.50      1st Qu.:13.00      1st Qu.: 6756      1st Qu.: 53.0      Yes: 69
## Median :13.70      Median :21.00      Median : 8418      Median : 65.0
## Mean   :14.13      Mean   :22.89      Mean   : 9689      Mean   : 65.5
## 3rd Qu.:16.50      3rd Qu.:31.00      3rd Qu.:10838      3rd Qu.: 78.0
## Max.   :39.80      Max.   :64.00      Max.   :56233      Max.   :100.0
```

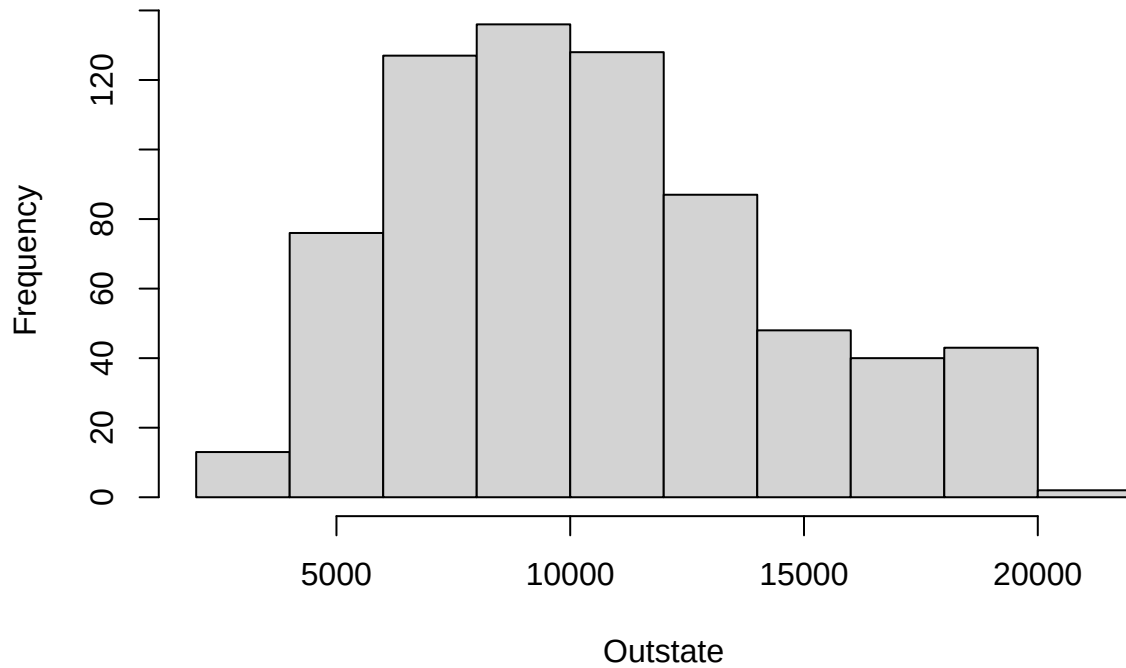
Number of private uni is 511 and public uni is 189.Elite universities are 631 and non Elite universities are 69.

1.3. Use mynewdata, plot histogram plots of four variables “Outstate”, “Room.Board”, “Books” and “Personal”. Give each plot a suitable title and label for x axis and y axis. (8%)

```
#setting x as Outstate,y as Room.Board,z as Books and i as Personal
x <- mynewdata$Outstate
y <- mynewdata$Room.Board
z <- mynewdata$Books
i <- mynewdata$Personal

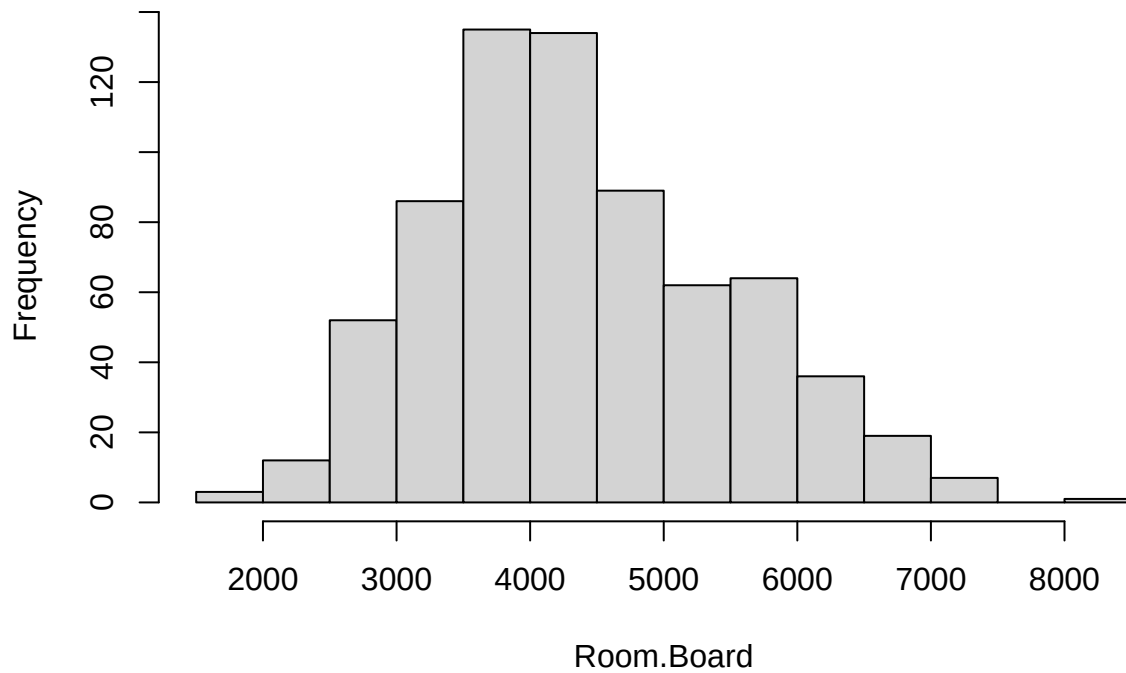
#Plotting Outstate histogram
hist(x, main = "Outstate histogram by frequency",xlab="Outstate",ylab = "Frequency")
```

Outstate histogram by frequency



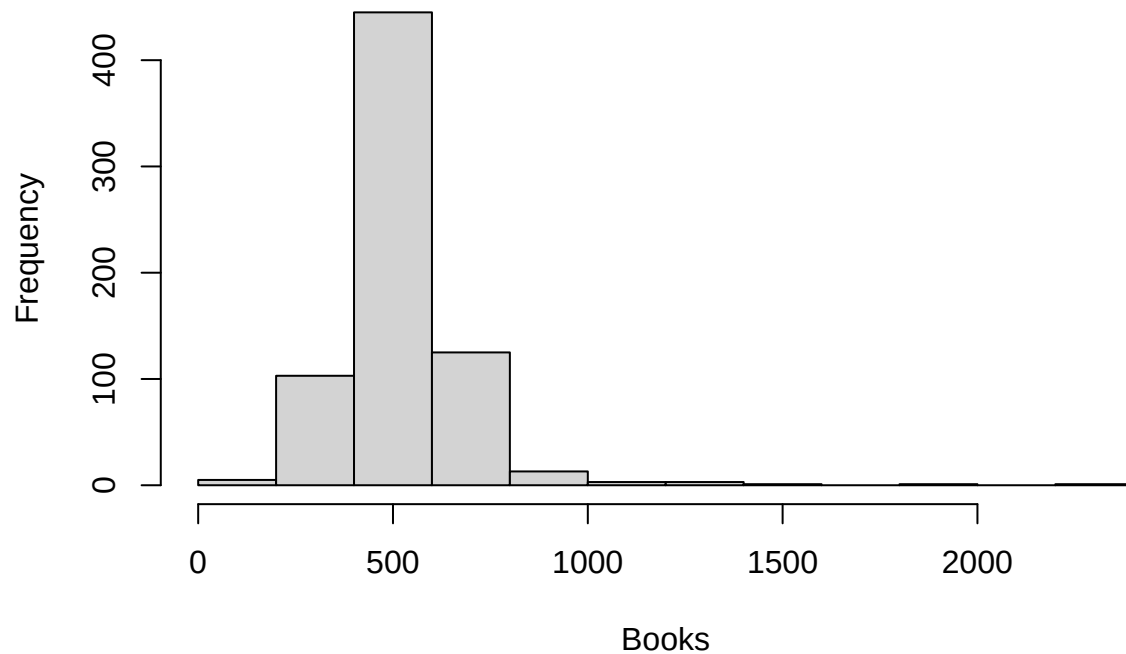
```
#Plotting Room board histogram  
hist(y, main = "Room board histogram by frequency",xlab="Room.Board",ylab = "Frequency")
```

Room board histogram by frequency



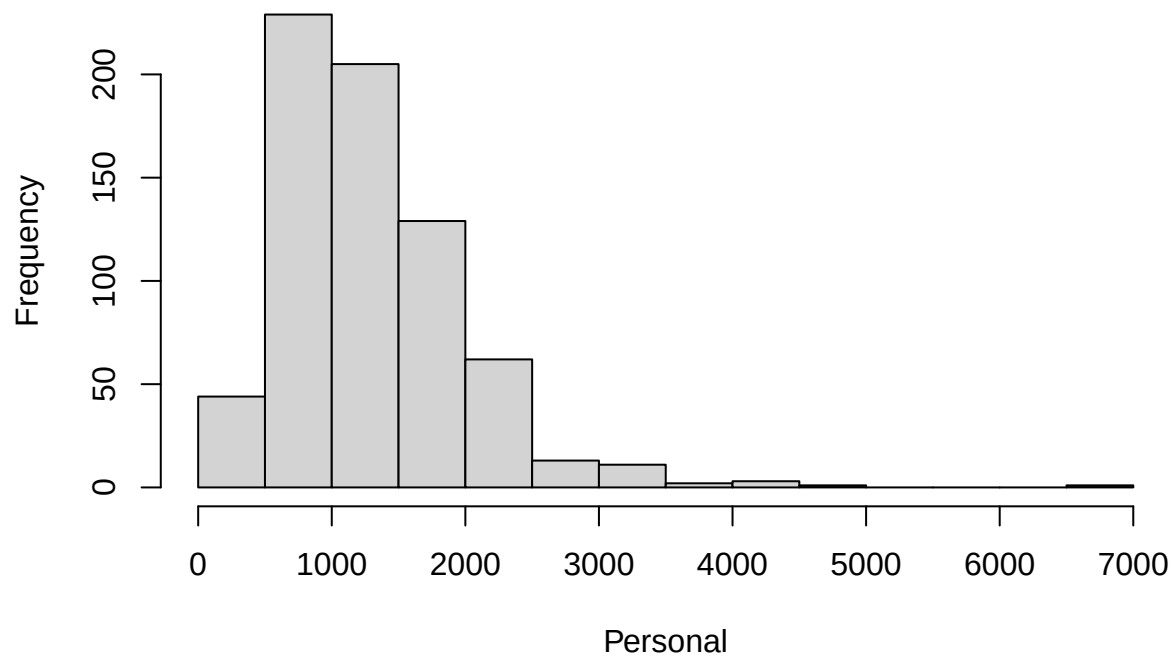
```
#Plotting Books histogram  
hist(z, main = "Books histogram by frequency",xlab="Books",ylab = "Frequency")
```

Books histogram by frequency



```
#Plotting Personal histogram  
hist(i, main = "Personal histogram by frequency", xlab="Personal", ylab = "Frequency")
```

Personal histogram by frequency



Task 2: Linear regression (45%)

2.1. Use mynewdata, do a linear regression fitting when outcome is “Grad.Rate” and predictors are “Private”

and “Elite”. Show the R output and report what you have learned from this output (you need to discuss significance, adjusted R-squared and p-value of F-statistics). (6%).

```
#using linear regression to fit the data and summary the output
myfit.sim = lm(Grad.Rate~Private+Elite ,data = mynewdata )
summary(myfit.sim)
```

```
##
## Call:
## lm(formula = Grad.Rate ~ Private + Elite, data = mynewdata)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -51.760 -10.150   0.545  10.240  44.850
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   55.150      1.109  49.724 <2e-16 ***
## PrivateYes    11.611      1.296   8.960 <2e-16 ***
## EliteYes      18.975      1.930   9.831 <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 15.17 on 697 degrees of freedom
## Multiple R-squared:  0.2166, Adjusted R-squared:  0.2144
## F-statistic: 96.38 on 2 and 697 DF,  p-value: < 2.2e-16
```

Private and Elite are statistically significant since the p-value is 2.2e-16 which is less than the alpha value 0.05.

Private and Elite explains about 22 percent of the variability of graduation rate

p-value of F test is smaller than 0.05, so the fitted model is better than null model.

2.2. Use the linear regression fitting result in 2.1, calculate the confidence intervals for the coefficients. Also give the prediction interval of “Grad.Rate” for a new data with Private=“Yes” and Elite=“No”. (4%)

```
#calculating confidence interval for the coefficient estimates
confint(myfit.sim)
```

```
##              2.5 %    97.5 %
## (Intercept) 52.971991 57.32724
## PrivateYes   9.066328 14.15484
## EliteYes    15.185492 22.76416
```

```
#prediction interval
predict(myfit.sim, newdata = data.frame(Private="Yes",Elite="No"),interval = "prediction")
```

```
##      fit      lwr      upr
## 1 66.7602 36.94453 96.57586
```

2.3 Use mynewdata, do a multiple linear regression fitting when outcome is “Grad.Rate”, all other variables as predictors. Show the R output and report what you have learned from this output (you need to discuss significance, adjusted R-squared and p-value of F-statistics). Is linear regression model in 2.3 better than linear regression in 2.1? Use ANOVA to justify your conclusion. (14%)

```
#all model, to include all other variables as predictors
myfit.all <-lm(Grad.Rate~.,data = mynewdata)
summary(myfit.all)
```

```
##
## Call:
## lm(formula = Grad.Rate ~ ., data = mynewdata)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -48.123  -7.138  -0.626   6.849  53.437
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 32.7705239  4.9097982   6.675 5.13e-11 ***
## PrivateYes   4.0721727  1.7757644   2.293 0.022140 *
## Apps         0.0016674  0.0004344   3.839 0.000135 ***
## Accept      -0.0013094  0.0008449  -1.550 0.121644
## Enroll       0.0024956  0.0023790   1.049 0.294534
## F.Undergrad -0.0003299  0.0004230  -0.780 0.435719
## P.Undergrad -0.0019171  0.0003995  -4.799 1.96e-06 ***
## Outstate     0.0010757  0.0002412   4.459 9.62e-06 ***
## Room.Board   0.0017412  0.0006075   2.866 0.004284 **
## Books        -0.0010752  0.0030147  -0.357 0.721462
## Personal     -0.0014240  0.0008006  -1.779 0.075741 .
## PhD          0.2005869  0.0585771   3.424 0.000653 ***
## Terminal    -0.0893928  0.0649373  -1.377 0.169086
## S.F.Ratio    0.0449358  0.1644932   0.273 0.784800
## perc.alumni  0.3169651  0.0501993   6.314 4.89e-10 ***
## Expend      -0.0003892  0.0001545  -2.519 0.012004 *
## EliteYes     3.1545935  2.1274226   1.483 0.138582
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 12.65 on 683 degrees of freedom
## Multiple R-squared:  0.4664, Adjusted R-squared:  0.4539
## F-statistic: 37.32 on 16 and 683 DF,  p-value: < 2.2e-16

#analysis of variance
anova(myfit.sim, myfit.all)

## Analysis of Variance Table
##
## Model 1: Grad.Rate ~ Private + Elite
## Model 2: Grad.Rate ~ Private + Apps + Accept + Enroll + F.Undergrad +
##          P.Undergrad + Outstate + Room.Board + Books + Personal +
##          PhD + Terminal + S.F.Ratio + perc.alumni + Expend + Elite
##   Res.Df    RSS Df Sum of Sq    F    Pr(>F)
## 1      697 160390
## 2      683 109246 14      51144 22.839 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

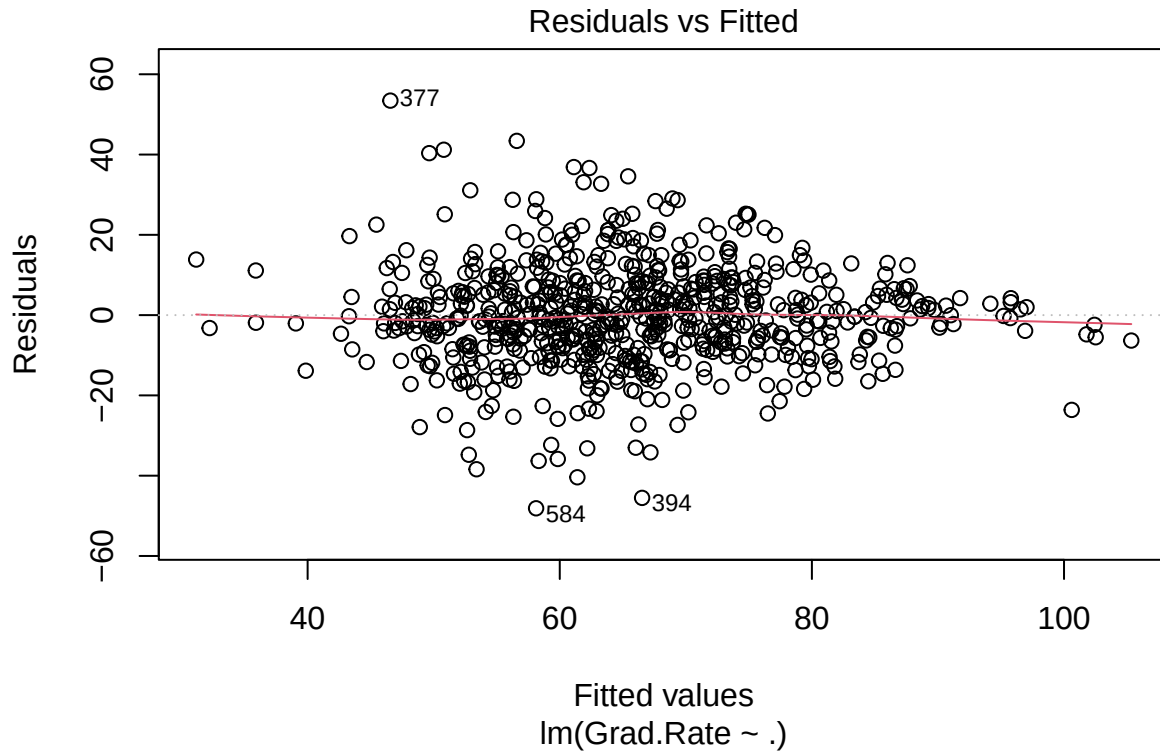
myfit.all explains about **45 percent** of the variance in graduation rate, which is roughly double the variance explained by the simple model with Elite and Private variables as the only predictors. myfit.all's **p-value of F test is smaller than 0.05**, so the all fitted model is better than null model. In myfit.all several variables are significant, this include Apps, P.Undergrad, Outstate, PhD, perc.alumni, Private, Room.Board. Other variables such as Enroll, Accept, F.Undergrad, Books, Terminal, Elite and S.F.Ratio are not significant.

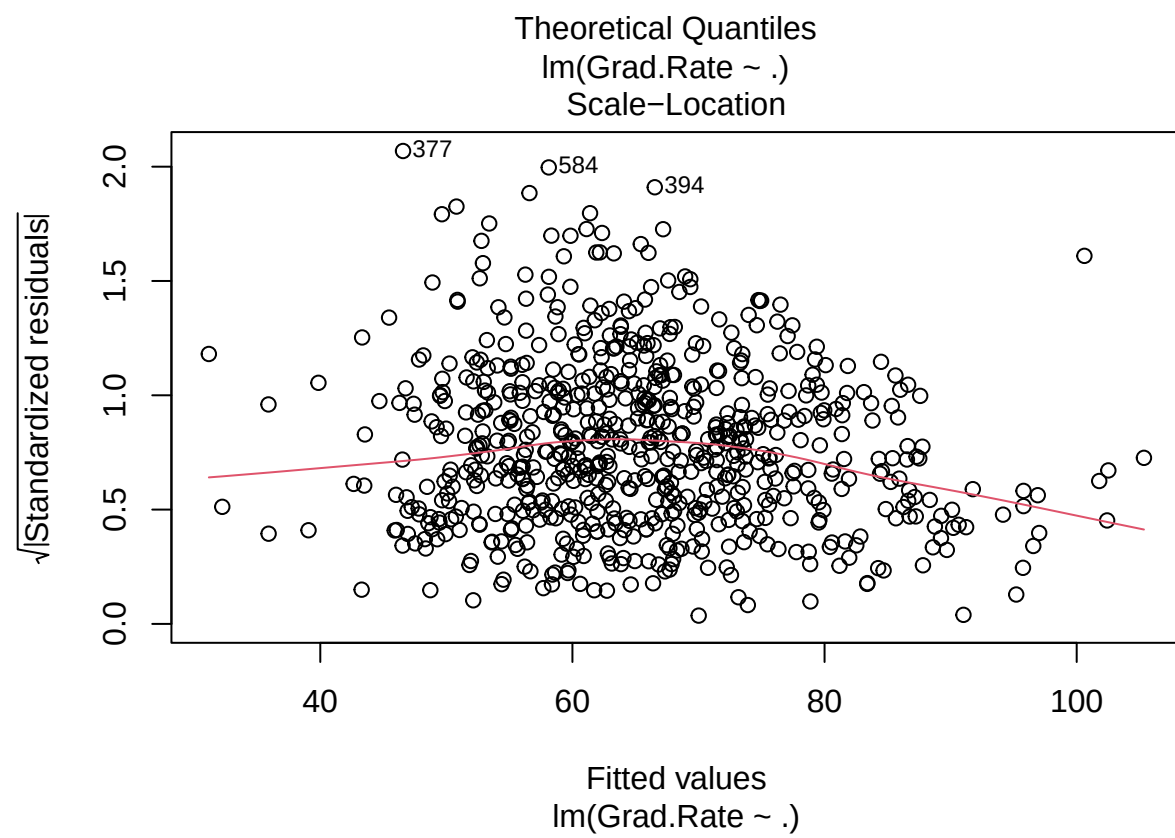
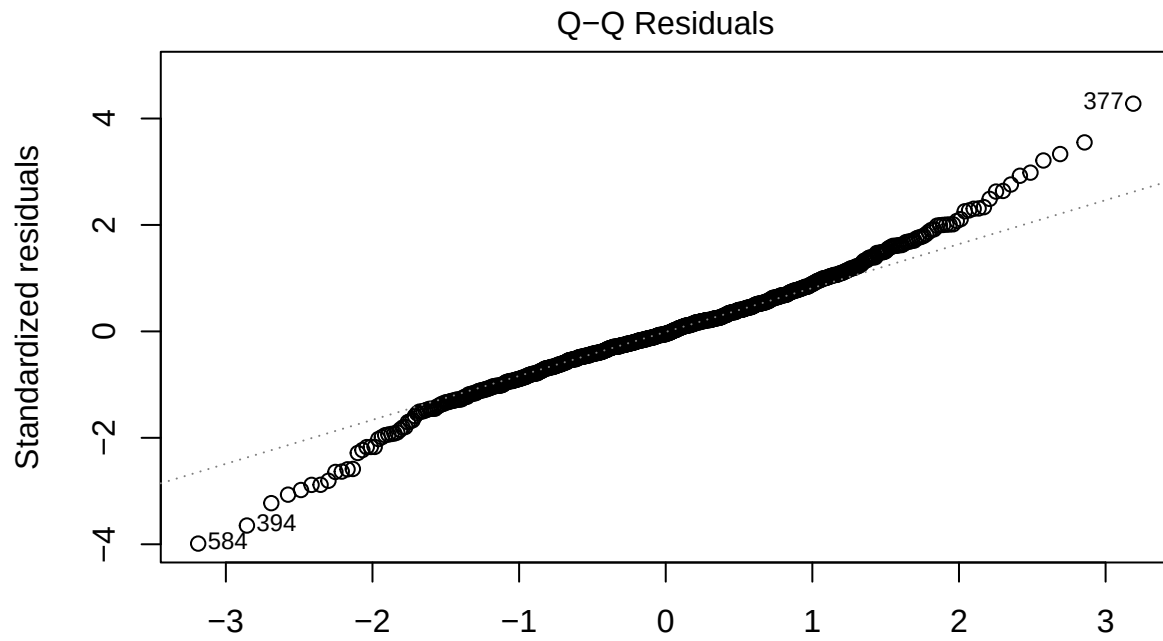
myfit.all (model 2) with all variables as predictors is **better** than myfit.sim(model 1) with only Elite and

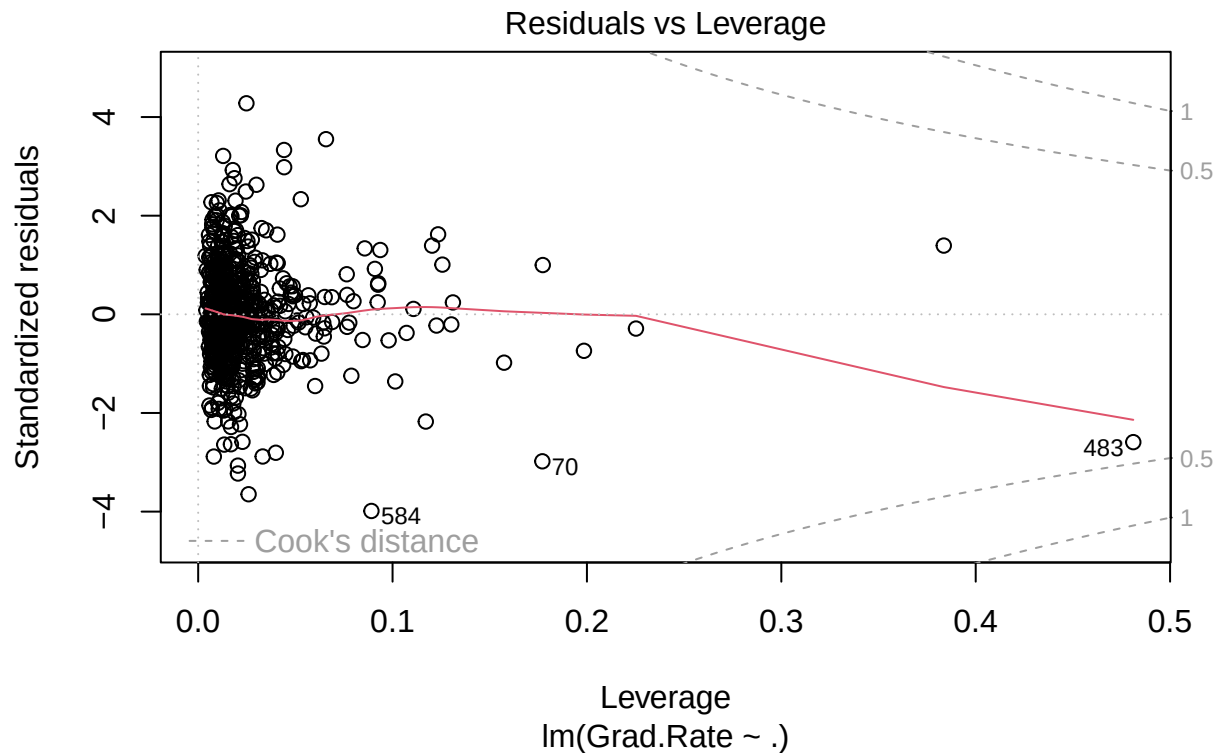
private as predictors. This is clear with the ANOVA comparison between them , showing a very small p_value of 2.2e-16 ,meaning the myfit.all is significantly better than myfit.sim.

2.4. Use the diagnostic plots to look at the fitting of multiple linear regression in 2.3. Please comment what you have seen from those plots. (7%)

```
#diagnostic plots  
plot(myfit.all)
```







Findings from diagnostics plot for myfit.all (model 2): The residual plot, the red smoothing fitting line appears to be slightly curved and we have outliers in our data. scale-Location plot, the red smoothing line is curved and this might suggest violation of homoscedasticity assumption. the QQ plot shows the normal distribution assumption for residuals is not fully satisfied and the leverage plot indicates there are some high influence (high leverage) points.

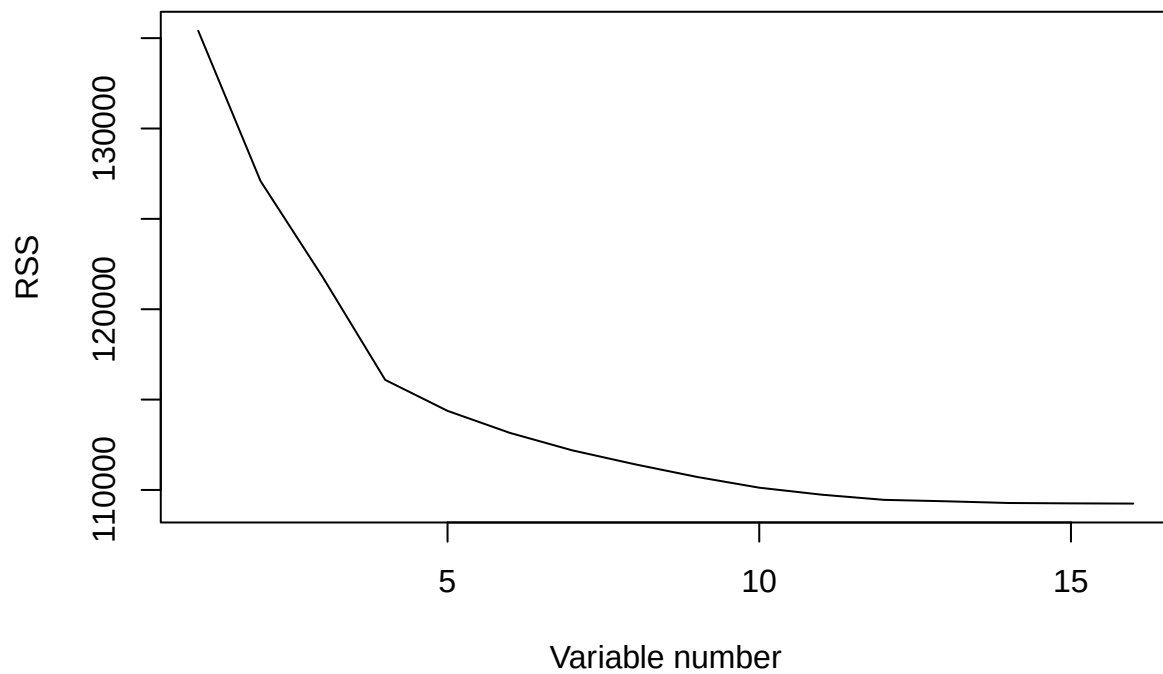
2.5. Use mynewdata, do a variable selection to choose the best model. You should use plots to justify how do you choose your best model. Use the selected predictors of your best model with outcome “Grad.Rate”, do a linear regression fitting and plot the diagnostic plots for this fitting. You can use either exhaustive, or forward, or backward selection method. (14%)

```
#model selection using backward selection method
```

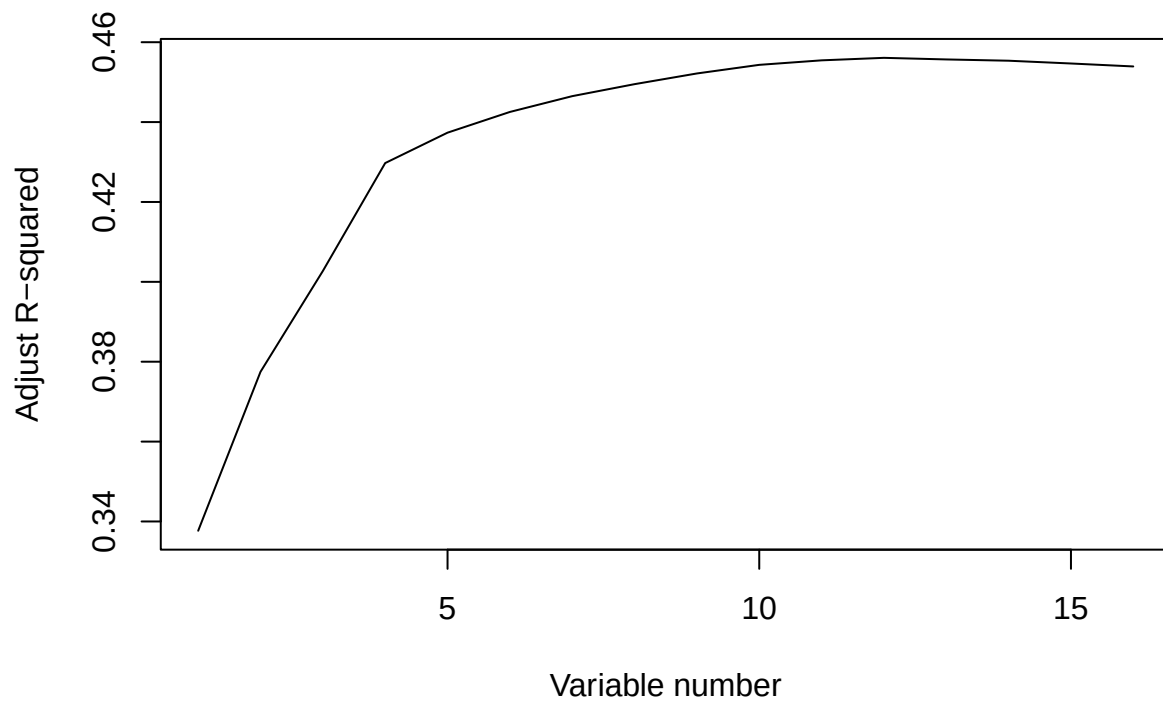
```
myfit.bwd<-regsubsets(Grad.Rate~., data=mynewdata, nvmax=16,method="backward")
myfit.bwd.sum<-summary(myfit.bwd)
```

```
#Plots to justify the decision of choosing the best model
```

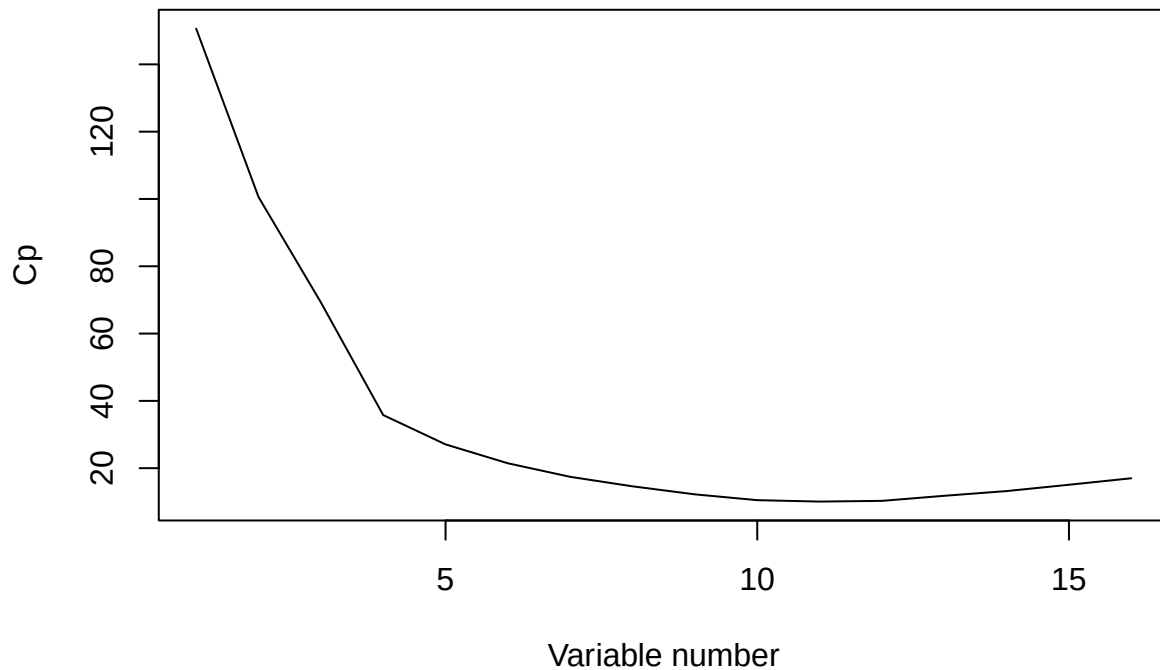
```
plot(myfit.bwd.sum$rss, xlab="Variable number", ylab="RSS", type="l")
```



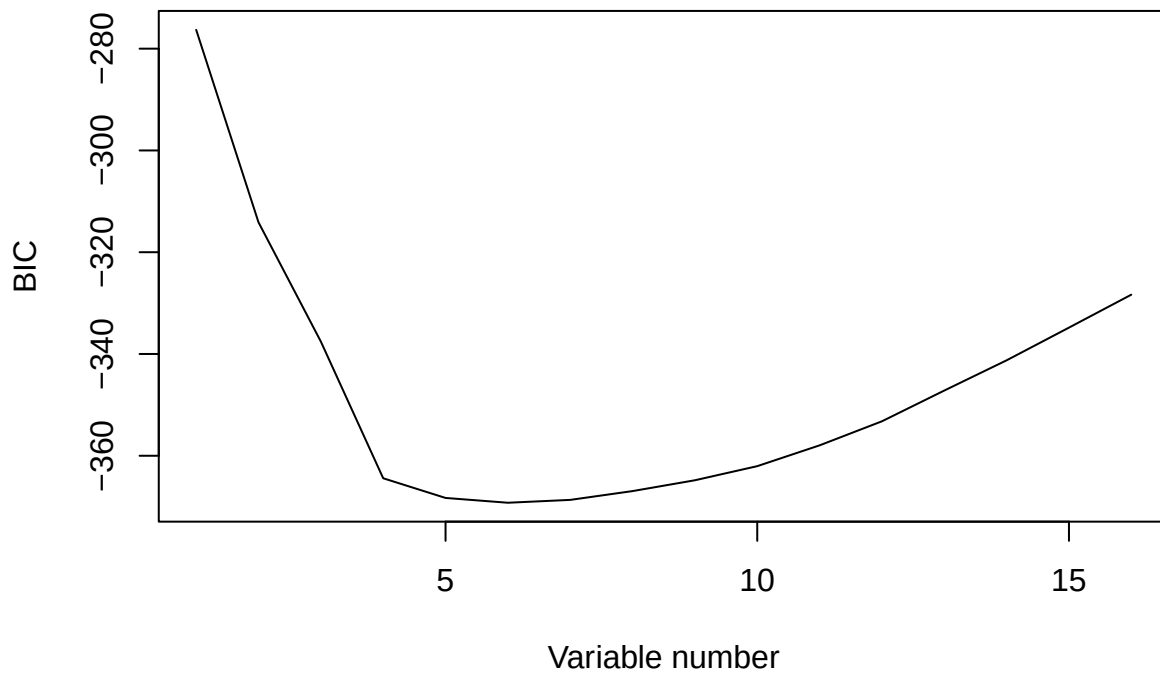
```
plot(myfit.bwd.sum$adjr2, xlab="Variable number", ylab="Adjust R-squared", type="l")
```



```
plot(myfit.bwd.sum$cp, xlab="Variable number", ylab="Cp", type="l")
```



```
plot(myfit.bwd.sum$bic, xlab="Variable number", ylab="BIC", type="l")
```



From
 plot RSS Cp and BIC are all decreasing and adjusted R-squared is increasing. We will use `which.min()` or `which.max()` to check the best model.

```
#check which model gives the maximum adjusted R-squared
which.max(myfit.bwd.sum$adjr2)
```

```
## [1] 12
```

```
#check which model gives the minimum cp
which.min(myfit.bwd.sum$cp)
```

```
## [1] 11
```

```
#check which model gives the minimum cp
which.min(myfit.bwd.sum$bic)
```

```
## [1] 6
```

```
which.min(myfit.bwd.sum$rss)
```

```
## [1] 16
```

Adjusted R-squared is the largest when variable number is 12, cp is the smallest when variable number is 11 and BIC is the smallest when variable number is 6. We will choose the model that uses the least number of variables. We will select model 6 by BIC.

```
#Looking at the predictors selected by BIC
coef(myfit.bwd, 6)
```

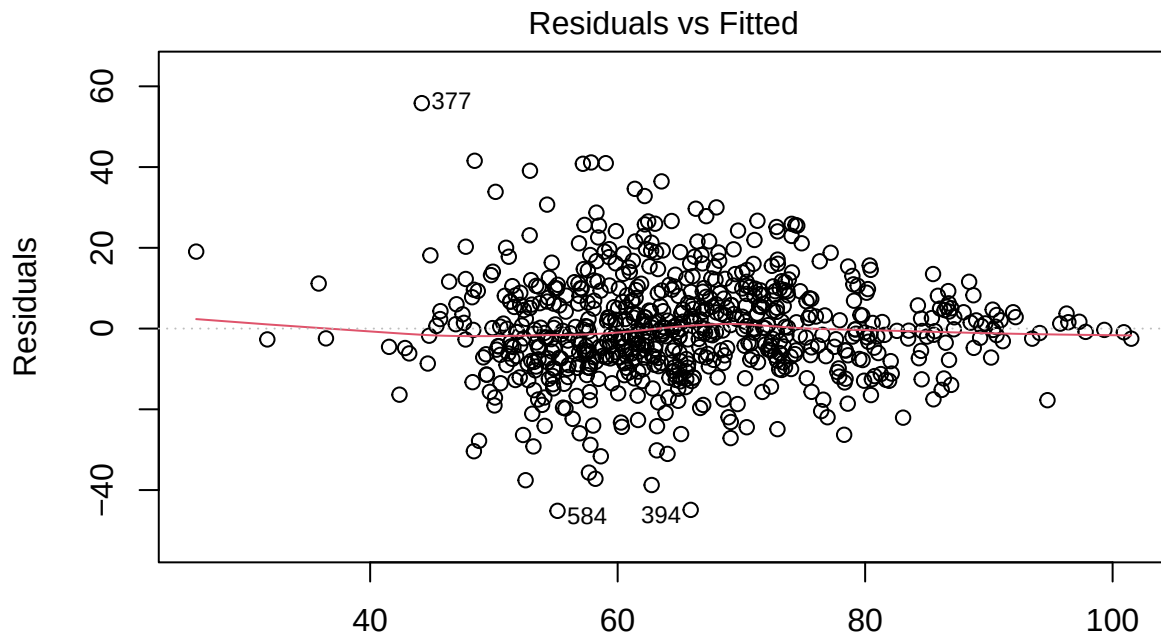
```
##      (Intercept)          Apps  P.Undergrad      Outstate    Room.Board
## 30.3835877456  0.0008506657 -0.0023796538  0.0011015386  0.0018146083
##           PhD      perc.alumni
##  0.0998980461  0.3424686297
```

The variables selected are Apps,P.Undergrad,Outstate,Room.Board, PhD and perc.alumni

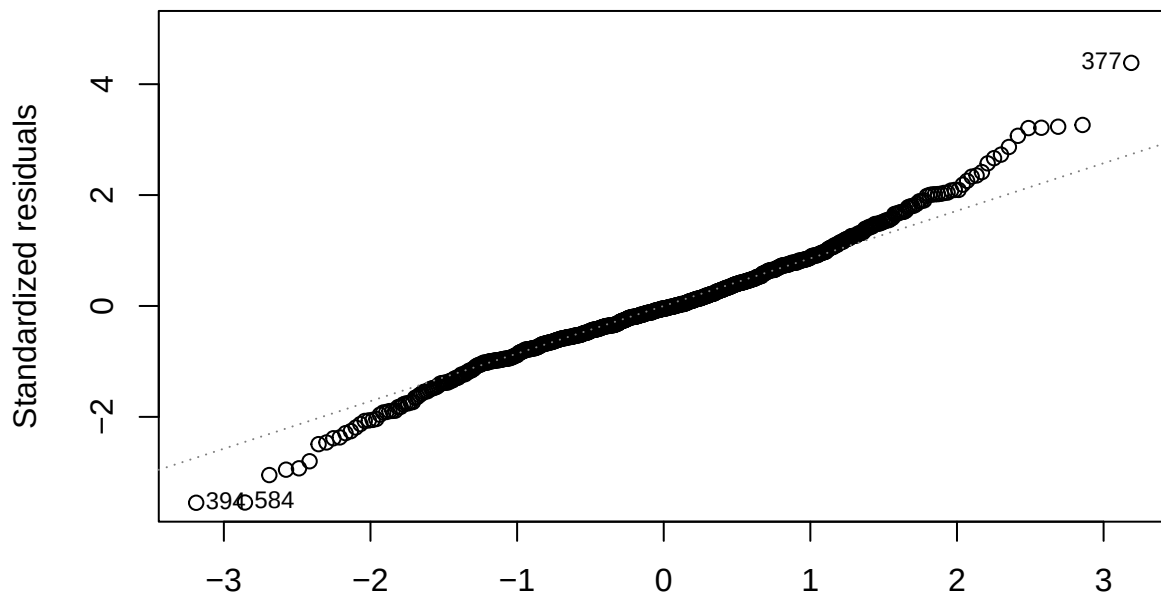
```
#using linear regression to fit Grad.Rate with the 6 variables selected
myfit.best = lm(Grad.Rate~Apps+P.Undergrad+Outstate+Room.Board+PhD+perc.alumni,data = mynewdata)
summary(myfit.best)
```

```
##
## Call:
## lm(formula = Grad.Rate ~ Apps + P.Undergrad + Outstate + Room.Board +
##      PhD + perc.alumni, data = mynewdata)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -45.141  -7.350  -0.591   7.366  55.829
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 30.3835877  2.6718711  11.372  < 2e-16 ***
## Apps         0.0008507  0.0001458   5.835 8.25e-09 ***
## P.Undergrad -0.0023797  0.0003694  -6.442 2.21e-10 ***
## Outstate     0.0011015  0.0001955   5.634 2.56e-08 ***
## Room.Board   0.0018146  0.0005892   3.080  0.00215 **
## PhD          0.0998980  0.0365784   2.731  0.00647 **
## perc.alumni  0.3424686  0.0486072   7.046 4.46e-12 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 12.78 on 693 degrees of freedom
## Multiple R-squared:  0.4473, Adjusted R-squared:  0.4425
## F-statistic: 93.48 on 6 and 693 DF, p-value: < 2.2e-16
```

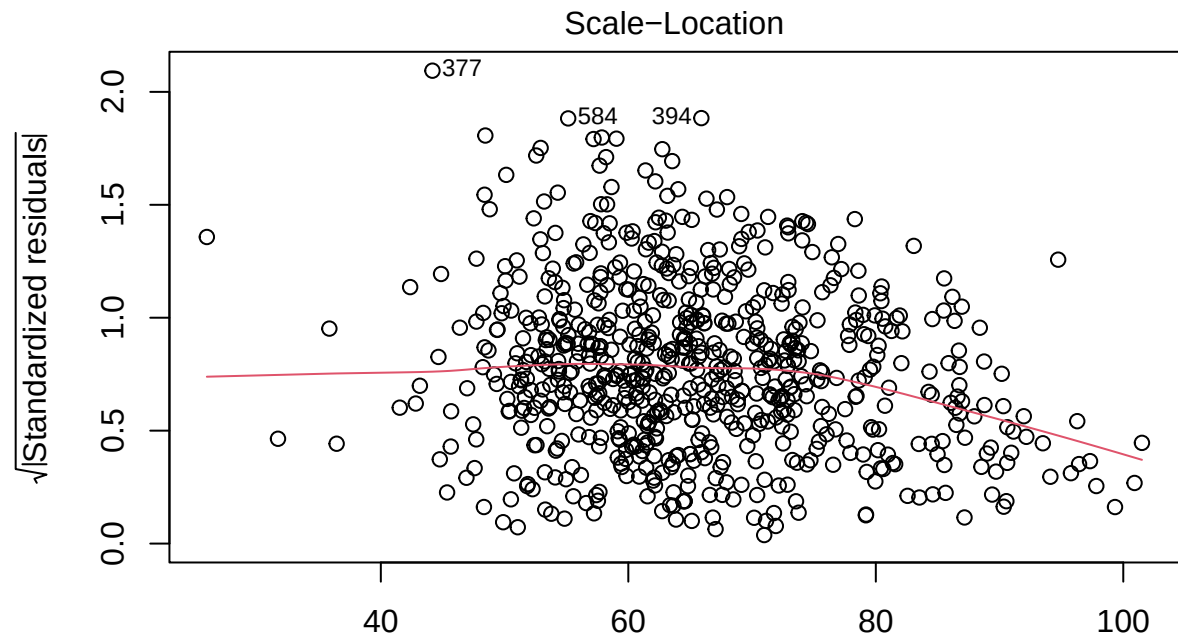
```
#diagnostic plots
plot(myfit.best)
```



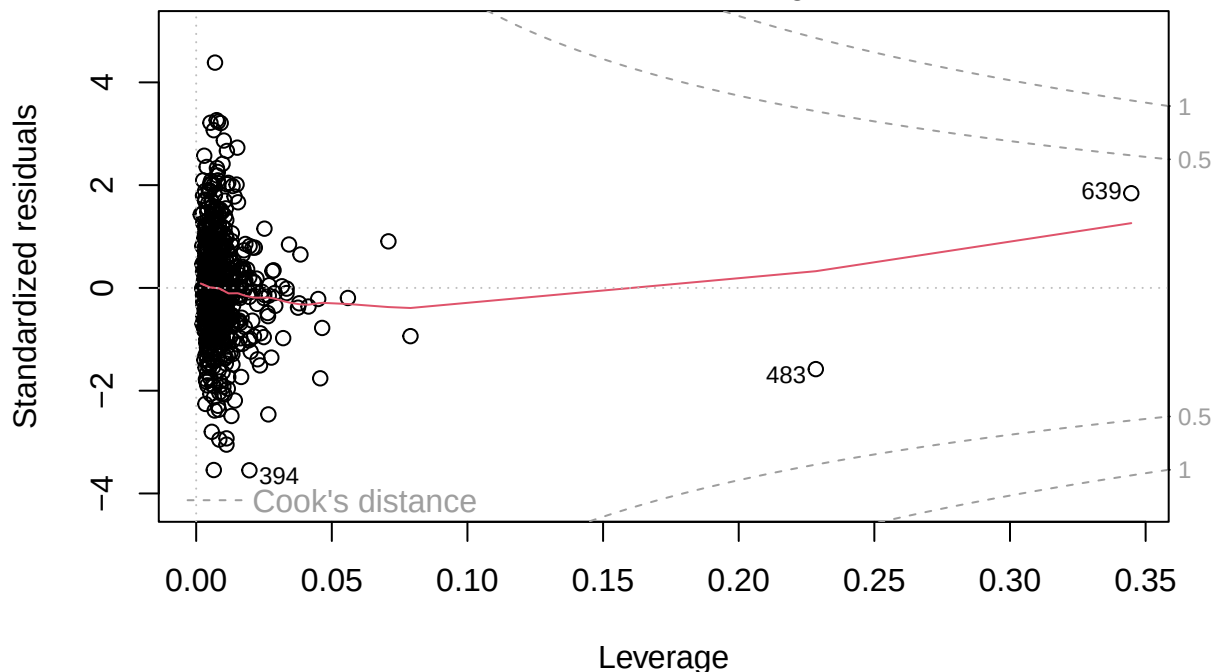
Im(Grad.Rate ~ Apps + P.Undergrad + Outstate + Room.Board + PhD + perc.alum .
Q-Q Residuals



Im(Grad.Rate ~ Apps + P.Undergrad + Outstate + Room.Board + PhD + perc.alum .



lm(Grad.Rate ~ Apps + P.Undergrad + Outstate + Room.Board + PhD + perc.alum .
Residuals vs Leverage



lm(Grad.Rate ~ Apps + P.Undergrad + Outstate + Room.Board + PhD + perc.alum .

Findings from diagnostics plot for myfit.best (model 3): The plots are almost similar to the plot in myfit.all

Task 3: Open question (30%)

Use mynewdata, discuss and perform any step(s) that you think that can improve the fitting in Task 2. You need to illustrate your work by using the R codes, output and discussion.

The diagnostic plots suggested that there might be non linearity in our data, we are going to use polynomial

regression to capture the complex patterns and log transformations

#Exploring Degree-2 Polynomial Regression for model improvement

```
myfit.pol<-lm(Grad.Rate ~ poly(Apps, 2, raw=T)+poly(Outstate, 2, raw=T)+poly(P.Undergrad, 2, raw=T)
+poly(Room.Board, 2, raw=T)+poly(PhD, 2, raw=T)+poly(perc.alumni, 2, raw=T),data=mynewdata)
```

```
summary(myfit.pol)
```

```
##
```

```
## Call:
```

```
## lm(formula = Grad.Rate ~ poly(Apps, 2, raw = T) + poly(Outstate,
##      2, raw = T) + poly(P.Undergrad, 2, raw = T) + poly(Room.Board,
##      2, raw = T) + poly(PhD, 2, raw = T) + poly(perc.alumni, 2,
##      raw = T), data = mynewdata)
##
```

```
## Residuals:
```

```
##      Min       1Q   Median       3Q      Max
## -45.577  -7.263  -0.451   7.296  59.241
##
```

```
## Coefficients:
```

```
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      2.413e+01  8.708e+00   2.771 0.005745 **
## poly(Apps, 2, raw = T)1    1.458e-03  2.568e-04   5.678 2.02e-08 ***
## poly(Apps, 2, raw = T)2   -2.118e-08  8.527e-09  -2.484 0.013244 *
## poly(Outstate, 2, raw = T)1  2.373e-03  7.112e-04   3.337 0.000893 ***
## poly(Outstate, 2, raw = T)2 -5.659e-08  2.991e-08  -1.892 0.058854 .
## poly(P.Undergrad, 2, raw = T)1 -3.462e-03  6.819e-04  -5.077 4.95e-07 ***
## poly(P.Undergrad, 2, raw = T)2  7.590e-08  4.493e-08   1.690 0.091576 .
## poly(Room.Board, 2, raw = T)1 -1.213e-03  3.217e-03  -0.377 0.706259
## poly(Room.Board, 2, raw = T)2  3.166e-07  3.357e-07   0.943 0.346094
## poly(PhD, 2, raw = T)1      2.442e-01  1.838e-01   1.328 0.184461
## poly(PhD, 2, raw = T)2     -1.136e-03  1.427e-03  -0.796 0.426294
## poly(perc.alumni, 2, raw = T)1  5.480e-01  1.507e-01   3.636 0.000297 ***
## poly(perc.alumni, 2, raw = T)2 -3.885e-03  2.659e-03  -1.461 0.144436
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
```

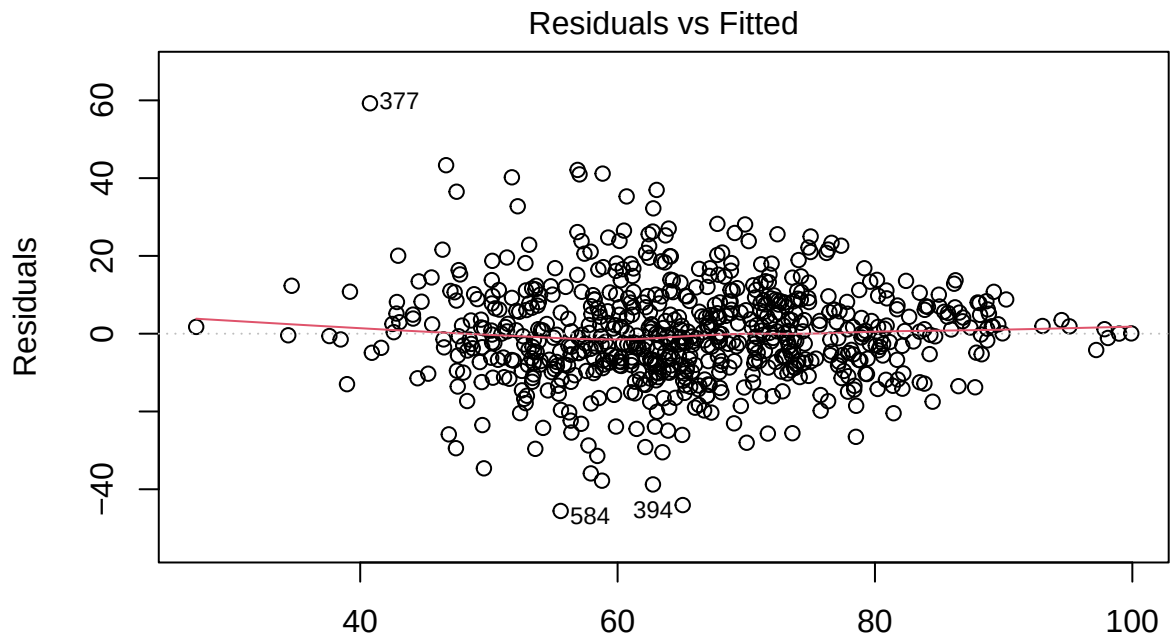
```
## Residual standard error: 12.68 on 687 degrees of freedom
```

```
## Multiple R-squared:  0.4607, Adjusted R-squared:  0.4513
```

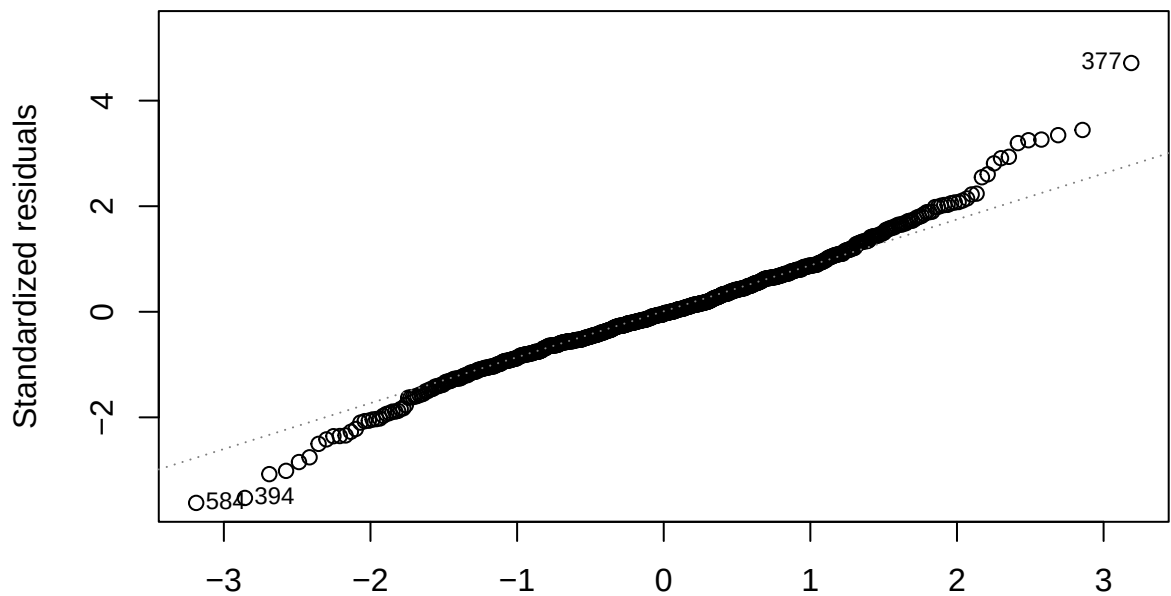
```
## F-statistic: 48.9 on 12 and 687 DF, p-value: < 2.2e-16
```

#diagnostic plots for myfit.pol

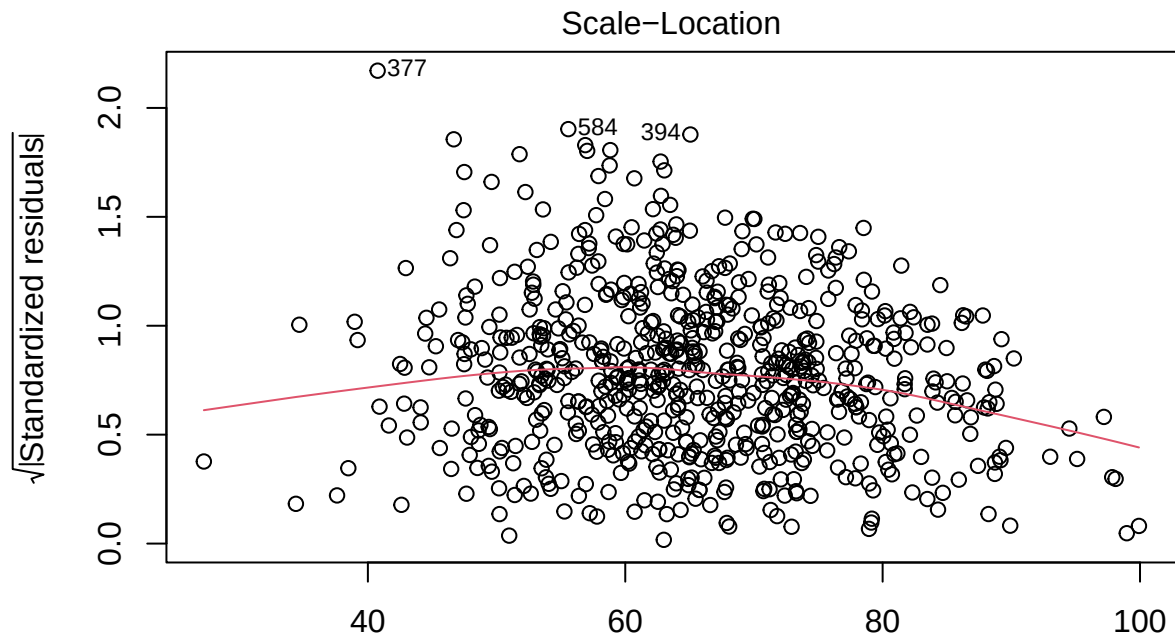
```
plot(myfit.pol)
```



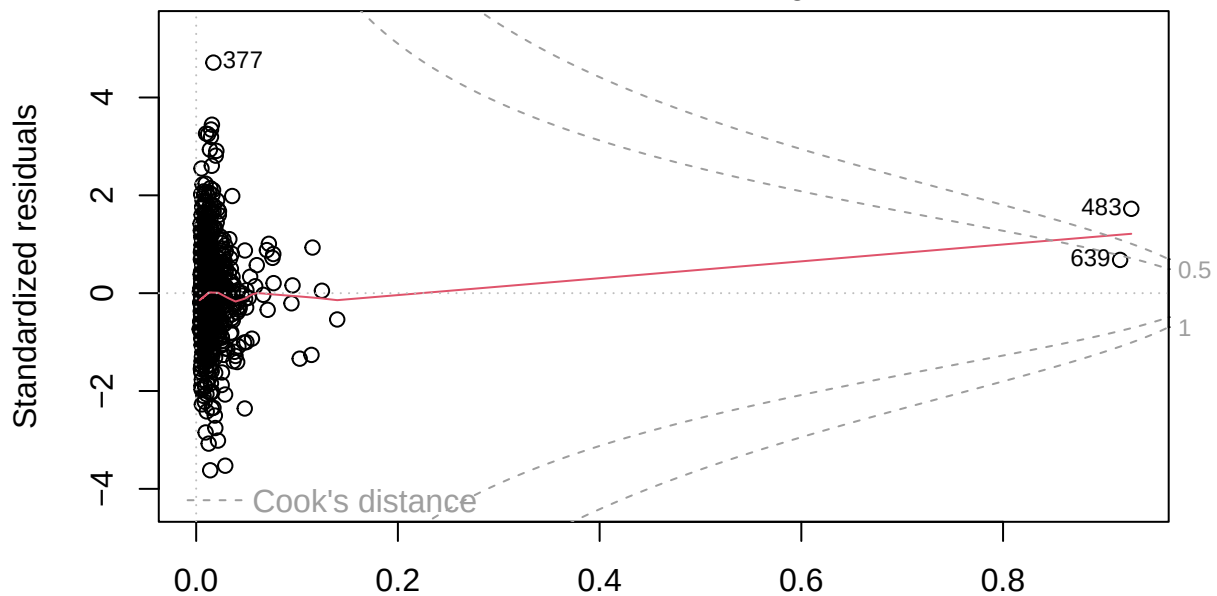
Fitted values
 $\text{lm}(\text{Grad.Rate} \sim \text{poly}(\text{Apps}, 2, \text{raw} = \text{T}) + \text{poly}(\text{Outstate}, 2, \text{raw} = \text{T}) + \text{poly}(\text{P} \dots$
 Q-Q Residuals



Theoretical Quantiles
 $\text{lm}(\text{Grad.Rate} \sim \text{poly}(\text{Apps}, 2, \text{raw} = \text{T}) + \text{poly}(\text{Outstate}, 2, \text{raw} = \text{T}) + \text{poly}(\text{P} \dots$



Im(Grad.Rate ~ poly(Apps, 2, raw = T) + poly(Outstate, 2, raw = T) + poly(P ...
Residuals vs Leverage



Im(Grad.Rate ~ poly(Apps, 2, raw = T) + poly(Outstate, 2, raw = T) + poly(P ... No

improvement

```
#anova between myfit.all and myfit.pol
anova(myfit.all, myfit.pol)
```

```
## Analysis of Variance Table
```

```
##
```

```
## Model 1: Grad.Rate ~ Private + Apps + Accept + Enroll + F.Undergrad +
```

```
##      P.Undergrad + Outstate + Room.Board + Books + Personal +
##      PhD + Terminal + S.F.Ratio + perc.alumni + Expend + Elite
## Model 2: Grad.Rate ~ poly(Apps, 2, raw = T) + poly(Outstate, 2, raw = T) +
##      poly(P.Undergrad, 2, raw = T) + poly(Room.Board, 2, raw = T) +
##      poly(PhD, 2, raw = T) + poly(perc.alumni, 2, raw = T)
## Res.Df    RSS Df Sum of Sq    F Pr(>F)
## 1      683 109246
## 2      687 110424 -4    -1178.6 1.8422  0.119
```

Comparing polynomial model and the multiple linear regression model, there is no significant improvement in the model.

we are going to mix polynomial and linear regression and see if it will improve our model.

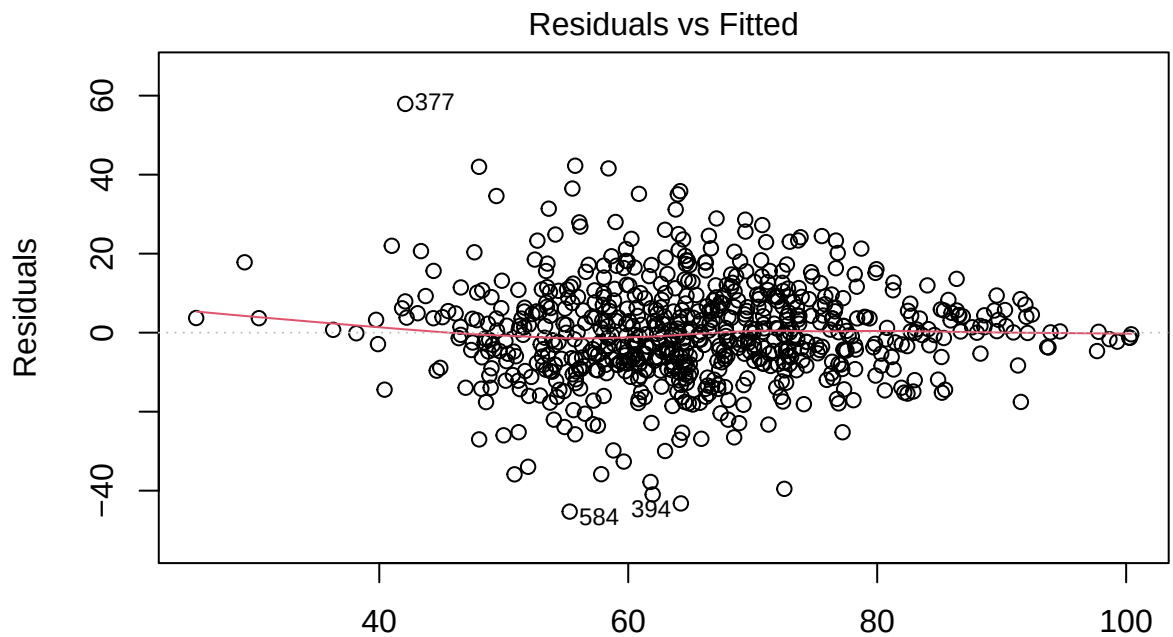
```
# Mixed model with polynomial for some predictors and linear for some
myfit.mixed <- lm(Grad.Rate ~ poly(Apps, 3, raw = TRUE) +
                  Outstate +
                  poly(P.Undergrad, 3, raw = TRUE) +
                  Room.Board +
                  PhD +
                  poly(perc.alumni, 2, raw = TRUE) +
                  Private + Elite,
                  data = mynewdata)

summary(myfit.mixed)

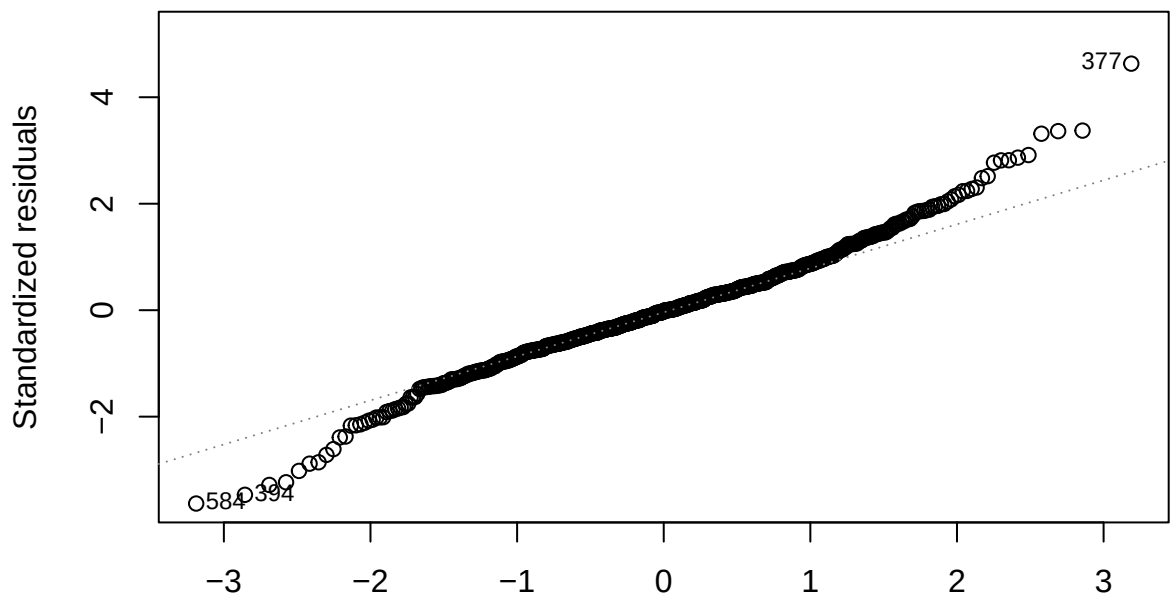
##
## Call:
## lm(formula = Grad.Rate ~ poly(Apps, 3, raw = TRUE) + Outstate +
##      poly(P.Undergrad, 3, raw = TRUE) + Room.Board + PhD + poly(perc.alumni,
##      2, raw = TRUE) + Private + Elite, data = mynewdata)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -45.297  -7.381  -0.239   6.488  57.908
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    2.560e+01  3.180e+00  8.051 3.64e-15 ***
## poly(Apps, 3, raw = TRUE)1    2.473e-03  5.690e-04  4.346 1.59e-05 ***
## poly(Apps, 3, raw = TRUE)2   -1.107e-07  4.579e-08 -2.418 0.015867 *
## poly(Apps, 3, raw = TRUE)3    1.501e-12  7.629e-13  1.967 0.049549 *
## Outstate        7.379e-04  2.228e-04  3.311 0.000977 ***
## poly(P.Undergrad, 3, raw = TRUE)1 -1.503e-03  1.266e-03 -1.187 0.235582
## poly(P.Undergrad, 3, raw = TRUE)2 -2.716e-07  2.292e-07 -1.185 0.236484
## poly(P.Undergrad, 3, raw = TRUE)3  1.257e-11  8.507e-12  1.478 0.139880
## Room.Board      1.389e-03  5.970e-04  2.327 0.020277 *
## PhD            1.050e-01  3.905e-02  2.689 0.007344 **
## poly(perc.alumni, 2, raw = TRUE)1  6.149e-01  1.457e-01  4.221 2.76e-05 ***
## poly(perc.alumni, 2, raw = TRUE)2 -5.414e-03  2.557e-03 -2.117 0.034587 *
## PrivateYes      5.806e+00  1.816e+00  3.198 0.001447 **
## EliteYes        2.448e+00  1.952e+00  1.254 0.210272
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 12.6 on 686 degrees of freedom
```

```
## Multiple R-squared:  0.4683, Adjusted R-squared:  0.4582
## F-statistic: 46.48 on 13 and 686 DF,  p-value: < 2.2e-16
```

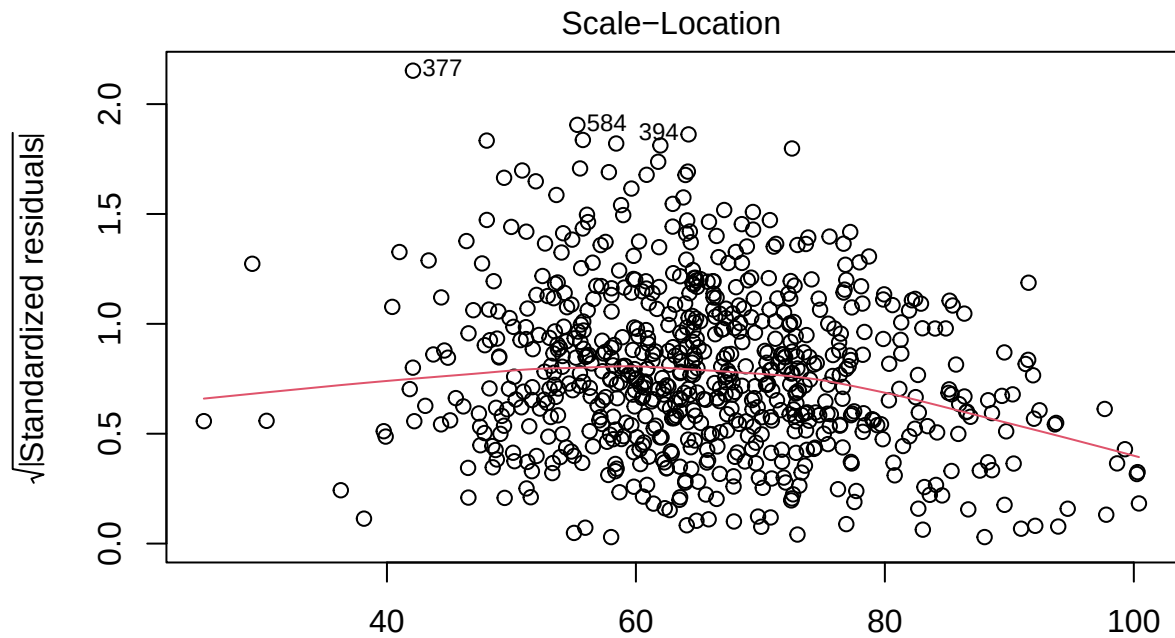
```
#diagnostic plots of mixed model
plot(myfit.mixed)
```



Fitted values
lm(Grad.Rate ~ poly(Apps, 3, raw = TRUE) + Outstate + poly(P.Undergrad, 3, ...
Q-Q Residuals

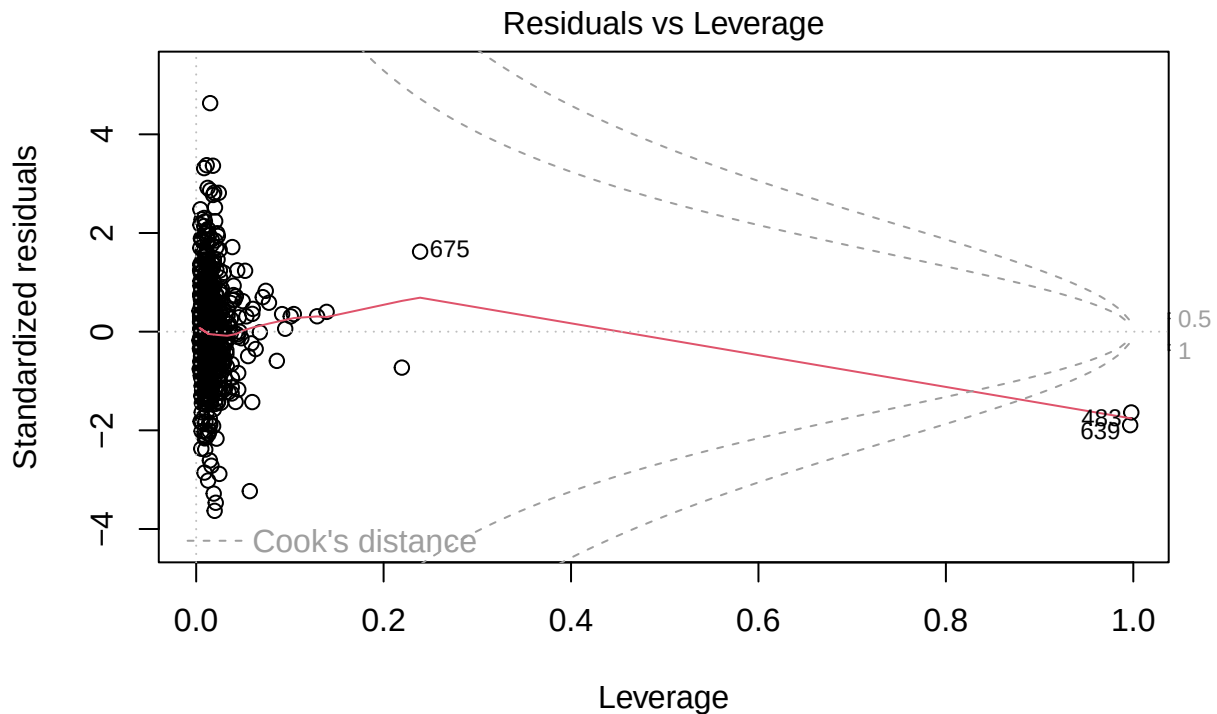


Theoretical Quantiles
lm(Grad.Rate ~ poly(Apps, 3, raw = TRUE) + Outstate + poly(P.Undergrad, 3, ...



`lm(Grad.Rate ~ poly(Apps, 3, raw = TRUE) + Outstate + poly(P.Undergrad, 3, ...`

```
## Warning in sqrt(crit * p * (1 - hh)/hh): NaNs produced
## Warning in sqrt(crit * p * (1 - hh)/hh): NaNs produced
```



`lm(Grad.Rate ~ poly(Apps, 3, raw = TRUE) + Outstate + poly(P.Undergrad, 3, ...`

slight improvement

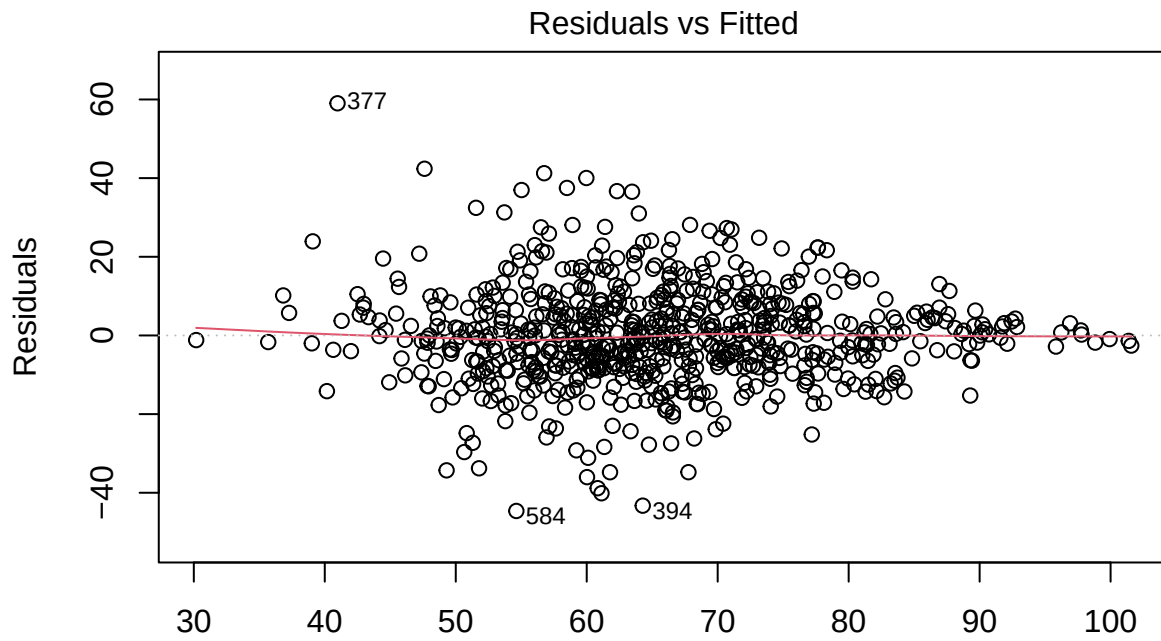
```
#anova of myfit.all and myfit.mixed models
anova(myfit.all, myfit.mixed)
```

```
## Analysis of Variance Table
##
## Model 1: Grad.Rate ~ Private + Apps + Accept + Enroll + F.Undergrad +
##      P.Undergrad + Outstate + Room.Board + Books + Personal +
##      PhD + Terminal + S.F.Ratio + perc.alumni + Expend + Elite
## Model 2: Grad.Rate ~ poly(Apps, 3, raw = TRUE) + Outstate + poly(P.Undergrad,
##      3, raw = TRUE) + Room.Board + PhD + poly(perc.alumni, 2,
##      raw = TRUE) + Private + Elite
##   Res.Df    RSS Df Sum of Sq F Pr(>F)
## 1      683 109246
## 2      686 108861 -3      384.67
```

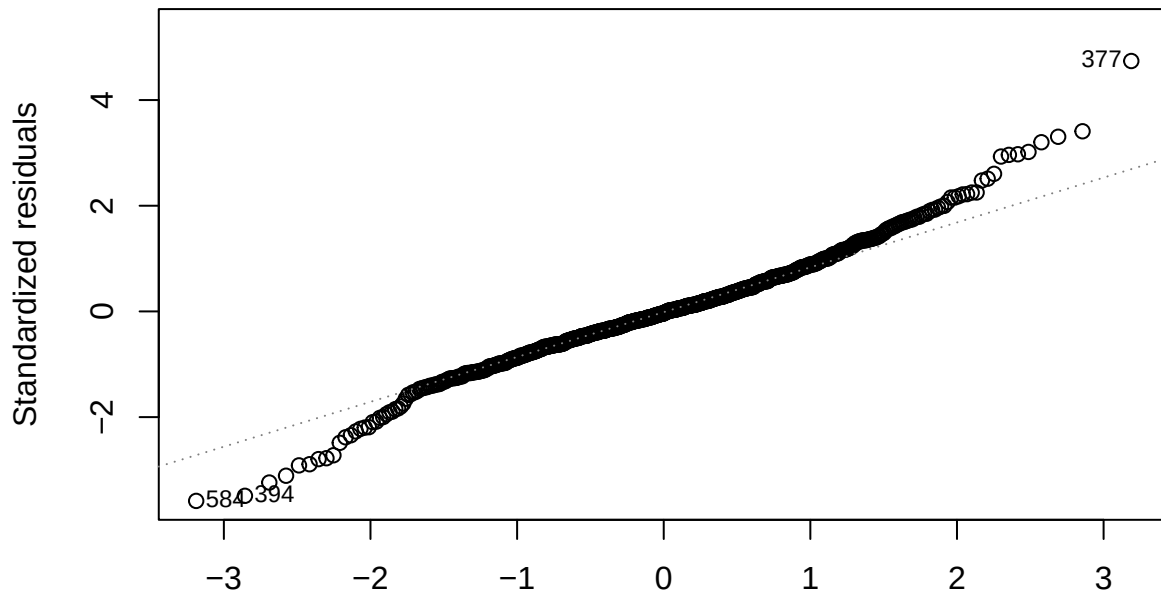
There is a slight improvement

```
#applying log transformation to Apps and Outstate to reduce skewness and fitting a polynomial regression
mynewdata$Apps <- log(mynewdata$Apps + 1)
mynewdata$Outstate <- log(mynewdata$Outstate + 1)
myfit.transformed <- lm(Grad.Rate ~ poly(Apps, 2, raw=TRUE) +
                        poly(P.Undergrad, 2, raw=TRUE) +
                        Outstate + Room.Board + PhD + perc.alumni +
                        Private + Elite, data=mynewdata)
summary(myfit.transformed)
```

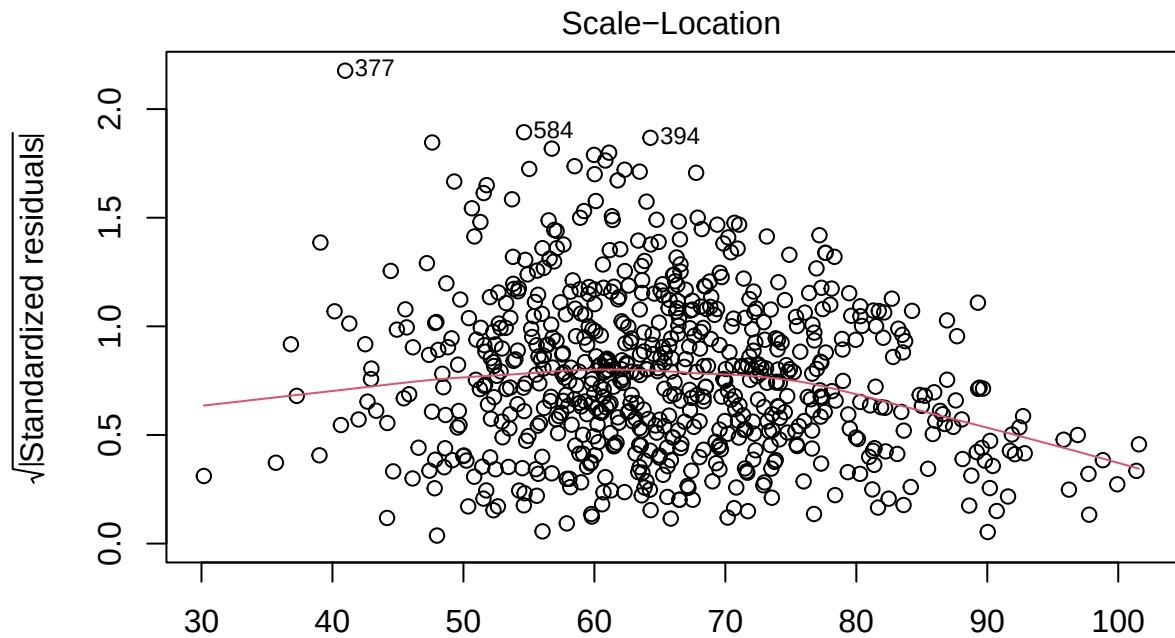
```
##
## Call:
## lm(formula = Grad.Rate ~ poly(Apps, 2, raw = TRUE) + poly(P.Undergrad,
##      2, raw = TRUE) + Outstate + Room.Board + PhD + perc.alumni +
##      Private + Elite, data = mynewdata)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -44.627  -7.255  -0.437   6.892  59.027
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    -2.861e+01  2.595e+01  -1.102  0.27066
## poly(Apps, 2, raw = TRUE)1    -3.138e+00  5.278e+00  -0.594  0.55238
## poly(Apps, 2, raw = TRUE)2     5.402e-01  3.548e-01   1.523  0.12832
## poly(P.Undergrad, 2, raw = TRUE)1 -3.259e-03  6.941e-04  -4.696 3.21e-06 ***
## poly(P.Undergrad, 2, raw = TRUE)2  7.101e-08  4.508e-08   1.575  0.11571
## Outstate        7.145e+00  2.119e+00   3.371  0.00079 ***
## Room.Board      1.427e-03  5.861e-04   2.435  0.01514 *
## PhD             8.169e-02  3.931e-02   2.078  0.03804 *
## perc.alumni     3.256e-01  4.843e-02   6.724 3.70e-11 ***
## PrivateYes      5.673e+00  1.845e+00   3.075  0.00219 **
## EliteYes        1.937e+00  1.879e+00   1.031  0.30282
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 12.54 on 689 degrees of freedom
## Multiple R-squared:  0.4705, Adjusted R-squared:  0.4628
## F-statistic: 61.21 on 10 and 689 DF, p-value: < 2.2e-16
#diagnostic plot of myfit.transformed
plot(myfit.transformed)
```



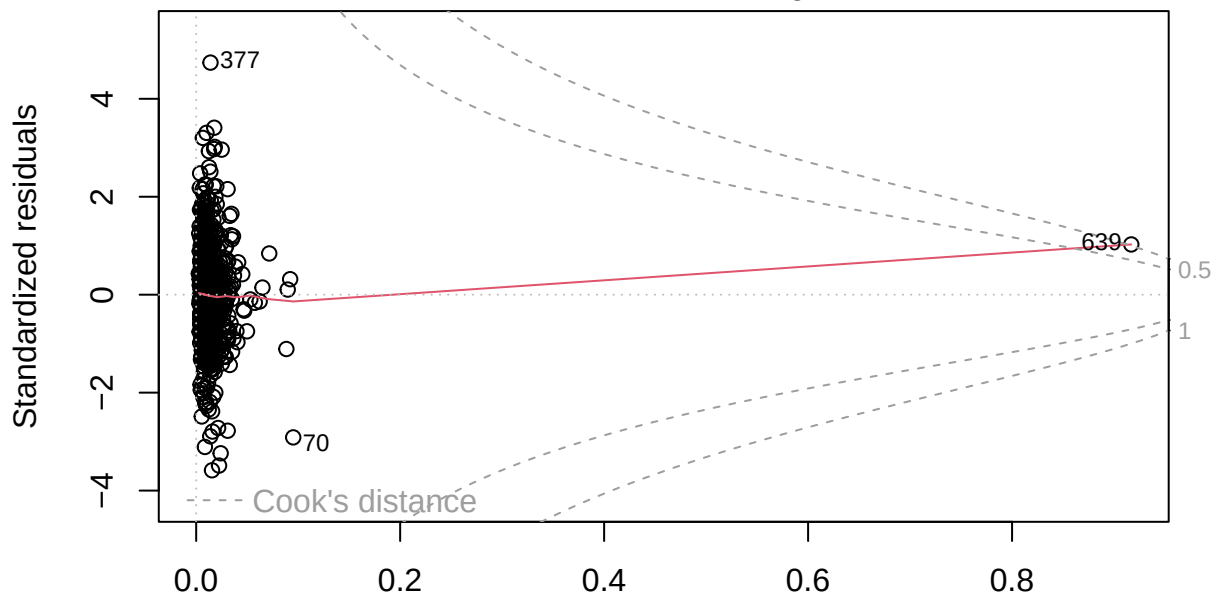
Fitted values
 $\text{lm}(\text{Grad.Rate} \sim \text{poly}(\text{Apps}, 2, \text{raw} = \text{TRUE}) + \text{poly}(\text{P.Undergrad}, 2, \text{raw} = \text{TRUE}) \dots$
 Q-Q Residuals



Theoretical Quantiles
 $\text{lm}(\text{Grad.Rate} \sim \text{poly}(\text{Apps}, 2, \text{raw} = \text{TRUE}) + \text{poly}(\text{P.Undergrad}, 2, \text{raw} = \text{TRUE}) \dots$



Residuals vs Leverage



Leverage

Model: $\text{lm}(\text{Grad.Rate} \sim \text{poly}(\text{Apps}, 2, \text{raw} = \text{TRUE}) + \text{poly}(\text{P.Undergrad}, 2, \text{raw} = \text{TRUE}) \dots$

slight improvement

```
#comparing myfit.all model and myfit.transformed model
anova(myfit.transformed, myfit.all)
```

```
## Analysis of Variance Table
##
## Model 1: Grad.Rate ~ poly(Apps, 2, raw = TRUE) + poly(P.Undergrad, 2,
```

```
##      raw = TRUE) + Outstate + Room.Board + PhD + perc.alumni +
##      Private + Elite
## Model 2: Grad.Rate ~ Private + Apps + Accept + Enroll + F.Undergrad +
##      P.Undergrad + Outstate + Room.Board + Books + Personal +
##      PhD + Terminal + S.F.Ratio + perc.alumni + Expend + Elite
## Res.Df    RSS Df Sum of Sq F Pr(>F)
## 1      689 108420
## 2      683 109246  6    -826.08
```

The myfit.transformed model is simpler and achieves a slightly higher adjusted R-squared, indicating a modest improvement in explanatory power. The diagnostic plots also suggest a minor improvement in fit. Additionally, the model is highly significant, with an F-statistic p-value $< 2.2\text{e-}16$.